



Chapter 6: Other Important Issues

(MDM-IS-EC302)

Lecture-2: AI and Innovation

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Other Important Issues

Ethical guidelines and responsible AI practices, **AI and Innovation**, Emerging trends and future directions in AI, AI and creativity: Generative models and artistic applications

What are AI Innovations?

- Advancements improving AI capabilities
- Focus: Intelligence, Automation, Interaction
- Driven by data, compute, algorithms
- **Evolution of AI**
 - Rule-based systems
 - Machine Learning
 - Deep Learning
 - Generative & Agentic AI

AI Innovations

- **AI in Healthcare**
 - Disease detection
 - Robotic surgery
 - Drug discovery
 - Personalized medicine
- **AI in Industry**
 - Smart manufacturing
 - Predictive maintenance
 - Supply chain optimization

AI Innovations

- **AI Agents**
 - Autonomous systems
 - Multi-agent collaboration
 - Decision-making systems
- **Multimodal AI**
 - Text + Image + Voice
 - Natural human interaction
 - Emotion recognition

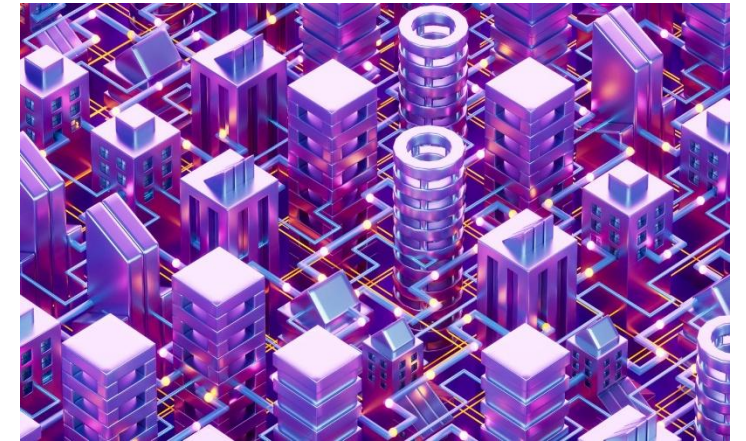
Future of AI

- Autonomous systems
- Smart cities
- Human-AI collaboration

AI Innovation: Relate Applications of AI in Daily Life

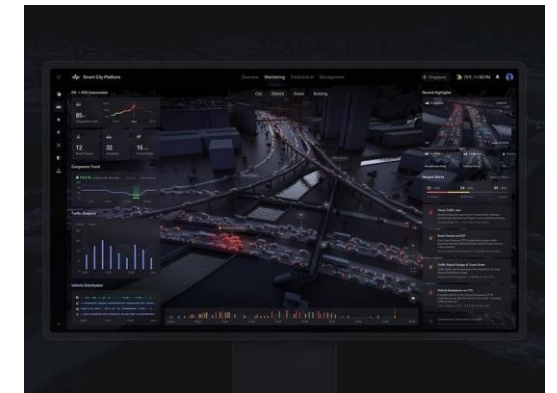
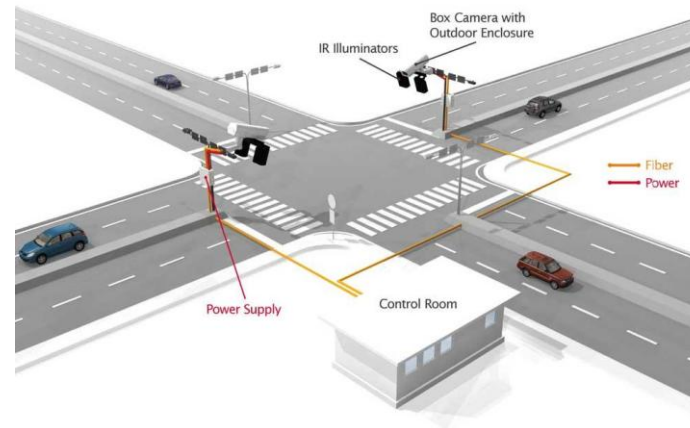
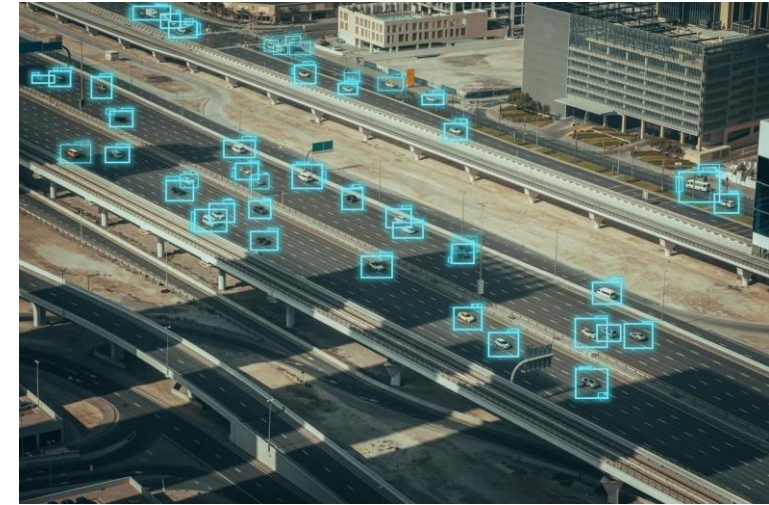
1. Smart Cities

- Building Intelligent, Sustainable Urban Ecosystems
- Urban area using digital technologies
- Integrates:
 - IoT + AI + Data Analytics
- Goal:
 - Improve quality of life
 - Efficient resource management



Role of AI in Smart Cities

- Data-driven decision making
- Real-time monitoring & automation
- Predictive analytics
- Intelligent governance
- Smart Traffic Management
 - AI-based traffic signal optimization
 - Congestion prediction
 - Accident detection
 - Smart parking systems



Role of AI in Smart Cities

- Smart Energy & Utilities
 - Smart grids & energy optimization
 - Demand forecasting
 - Renewable energy integration
 - Water & waste management



Role of AI in Smart Cities

- Public Safety & Surveillance
 - AI-based CCTV monitoring
 - Facial recognition
 - Crime prediction & prevention
 - Emergency response systems

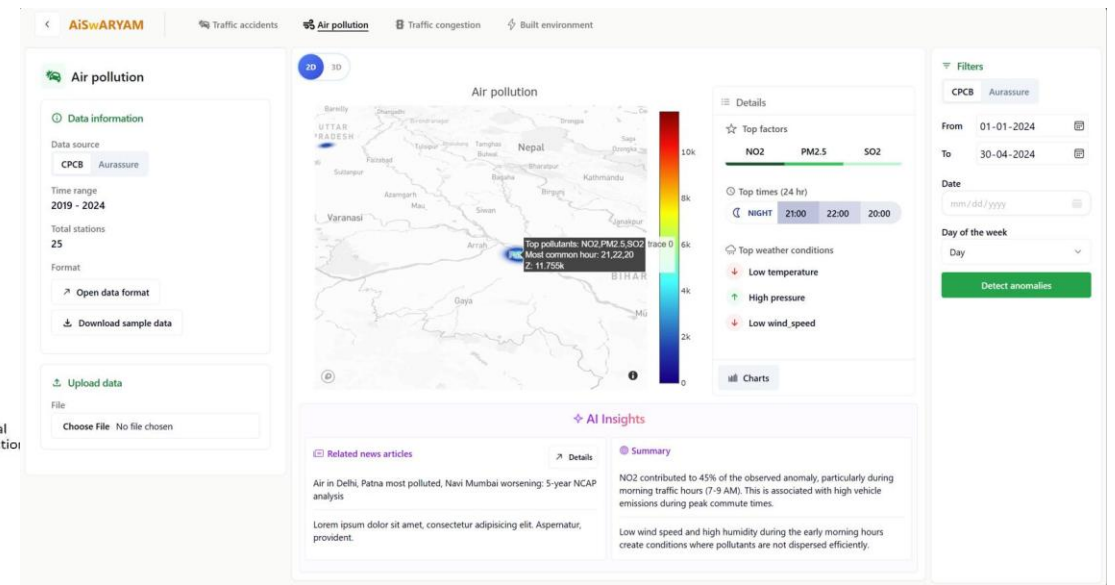
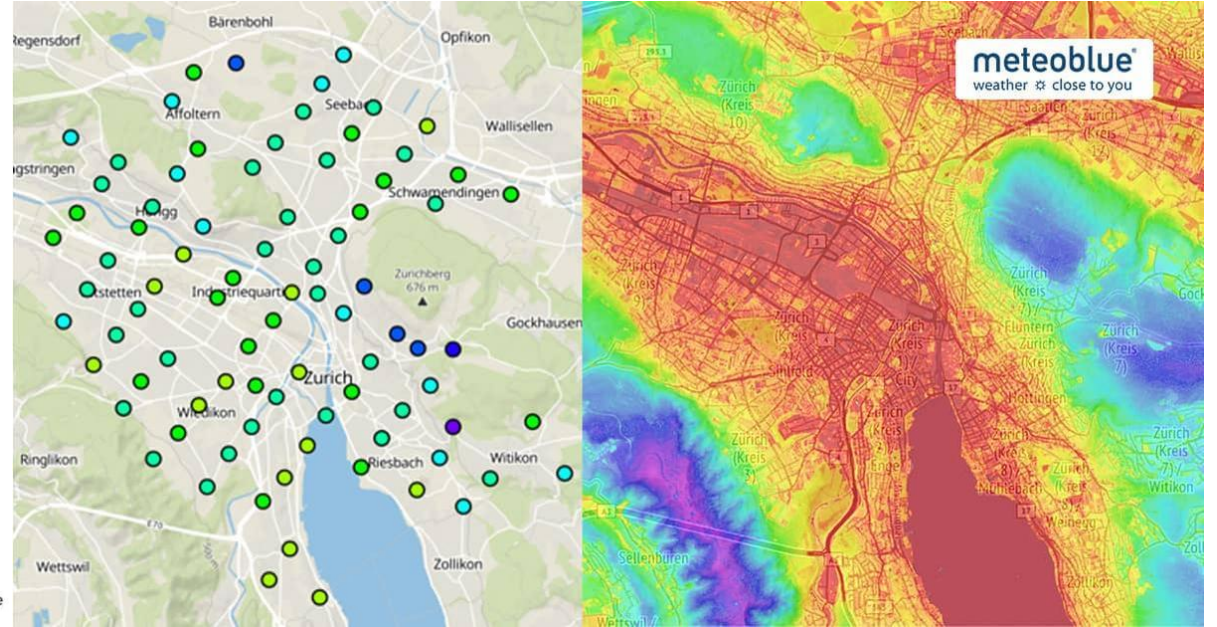
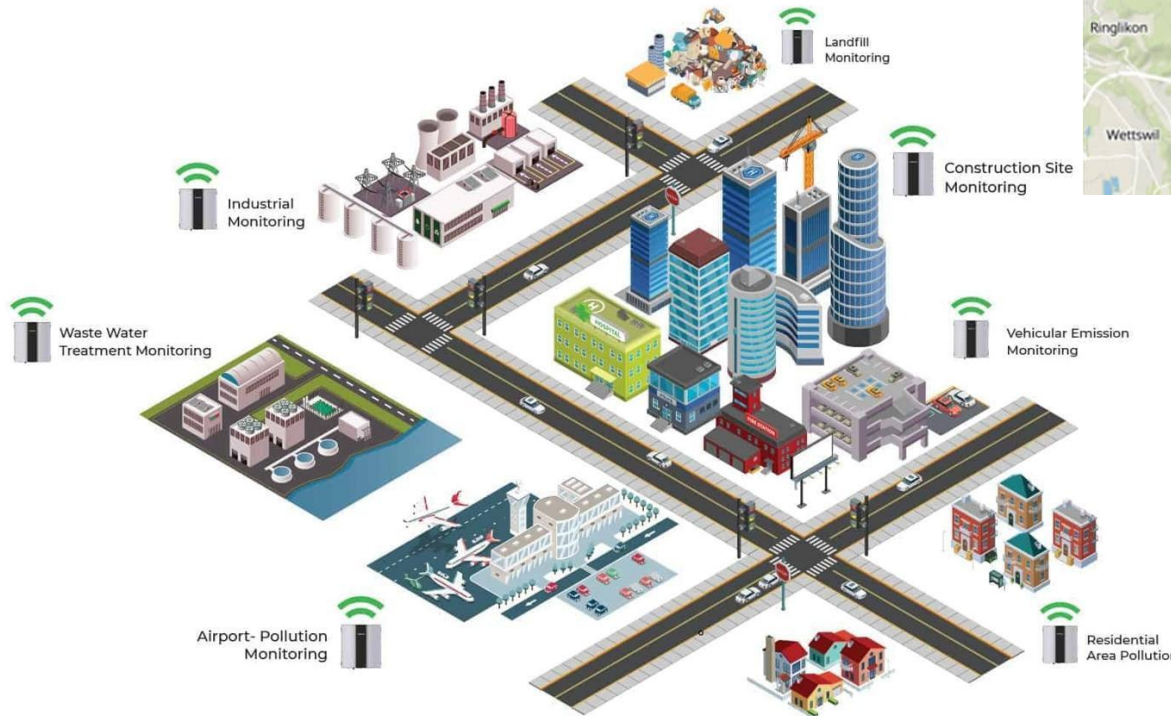


- Smart Healthcare Systems
 - Remote patient monitoring
 - AI diagnostics
 - Emergency healthcare optimization
 - Telemedicine services



Role of AI in Smart Cities

- Environmental Monitoring
 - Air & water quality tracking
 - Pollution prediction
 - Climate data analysis
 - Disaster management (floods, fires)



Future of Smart Cities

- Autonomous transport systems
- AI-powered governance
- Digital twins of cities
- Sustainable urban ecosystems



2. Smart Home

- The term “smart home technology” refers to any suite of appliances that can be independently and remotely controlled through a common network.
- For example, let’s assume that the thermostat, audio speakers, lights, security cameras, TV, etc., are **connected** to a network.
- As a part of smart home technology, it will be possible for you to control them from a single device – your smartphone.
- **Advantages Of Having A Smart Home**
 - **Benefit – 1: Managing Everything From A Single Location**
 - With a smart home, you can keep up with everything in your house by using a single device. In some cases, you may only need to install a single app to activate your security camera, ACs, TV, and other appliances.
 - **Benefit – 2: Better Energy Efficiency**
 - Considering how you’re using your smart home technology, it will also be possible for you to make everything much more energy efficient. For example, with these programs, you will have precise control over the heating or cooling system in your house.
 - **Benefit – 3: Maximize Home Security**
 - Incorporating your surveillance features with a smart home technology can also skyrocket your house’s overall security system.

3. Smart Schools

- A smart school more than just using advanced technologies for teaching and learning purposes
- **Advantages Of Having A Smart School**
- **Ability to Interact With Anyone Around Any Corner of the World**
 - The smart technology allows one to remotely interact with any person through collaboration via virtual learning platforms. Educational institutions can interact with students and manage a large user base without worrying about communication gaps.
- **Go ‘Green’ by Replacing Paper with Digital Tools**
 - The dynamic sharing of information and resources cuts down the need for paper, pen, printouts, etc. required for the learning process, and is the most vital advantage of a smart school.
- **Improved Productivity and Instant Access to Resources**
 - The visual model of the data shared through intelligent technology provides better engagement. Plenty of resources related to a single topic can be stored securely, and users can access every bit of them.

Smart building data connects to smart cities

Smart building data works in many areas, a few of which are as follows:

1. **Electricity:** In terms of electricity use, smart buildings monitor electricity demand and consumption in homes. They can reduce and save electricity based on family members' behavior. It can regulate power generation supply by helping power companies.
 2. **Maintenance:** The smart building data helps to provide better maintenance services and replacement.
 3. **Health:** It can also help to provide data on sick people in the building. Doctors can also provide monitoring to patients at their doorsteps.
- **Smart citizens**
 - Smart cities require smart citizens or may convert them into smart. Smart cities help governments achieve the Sustainable Development Goals. It can reduce waste, save water, and work with data on the poverty line, etc.
 - Smart citizens live life smartly, called smart living. Smart living is the next topic of Relate Applications of AI in Daily Life.
 - **Smart Living**
 - Smart living refers to using AI-enabled technologies in daily life. It involves gadgets that use such sensors and alerts to deal with environmental conditions. Closing and opening the doors, windows, etc., maintaining room temperature, etc.
 - In such critical situations like fire, gas leak, or any other emergencies, it can help by generating alerts.
 - Smart buildings can also generate danger alerts and warning systems in disasters. It also helps the government with disaster management.
 - It provides information alerts to the people at different level.

The impact of AI on Sustainable Development Goal (SDG)

The United Nations adopted 17 SDG Goals in the General Assembly in 2015. It was built upon the **"leaving no one behind"** principle.

- They were adopted for:
 1. Better Environment
 2. Better Society
 3. Better Economy
- The major challenges facing the world are providing clean water, clean air, natural resources, sustainable energy, and education.
- As we all know, we face many environmental problems, including poverty and starvation.
- So far, despite technological advances, we have not succeeded in solving these problems.
- So Artificial Intelligence can help to solve such problems.



- **No poverty (SDG - 1)**
- **Zero Hunger (SDG - 2)**
- **Good Health and Well Being (SDG - 3):** AI can help in preparing data for people living below the poverty line and help them by providing food, health, water, and energy easily.
- **Quality Education (SDG - 4):** AI systems can be trained for quality education and can provide the solution.
- **Gender Equality (SDG - 5):** A
 - AI tools don't recognize gender.
 - In some areas, AI can also result in negative impacts.
 - For example, AI systems or tools are trained based on the country's situation and requirements where they were developed.
 - If we talk about SDG 5 - Gender Equality, there is inefficient research done.
 - The tools available, such as voice agents and chatbots like Alexa or Siri, are gendered as female-only.
 - To improve this, more research is required for smart algorithms, image recognition, reinforcement learning, or discrimination.

- **Clean water and Sanitation (SDG - 6):** AI tools can be used for supplying clean water and sanitation to the people.
- **Affordable and Clean Energy (SDG - 7):** AI can be trained to provide affordable and clean energy, but the AI product design should require proper resources and tools.
- **Decent Work and Economic Growth (SDG - 8):** The market heavily relies on data analysis. These data may not be available in low and middle-income countries. Hence, the work culture may be impacted.
- **Industry Innovation and Infrastructure (SDG - 9):** An AI system helps in industry innovation in terms of solving complex problems. It can handle complex operations easily. It can also handle the infrastructural issues with advancement.
- **Reduce Inequalities (SDG – 10):** By using AI tools, inequalities can be reduced as AI tools do not have any emotions. They can produce better results than other tools. If we take the example of AI used by social media websites to show users' interests in their profiles. Another example is job creation using AI tools.
- **Sustainable Cities and Communities (SDG - 11):** AI is capable of creating smart cities that provide efficient resources to the people.

- **Responsible Consumption and Production (SDG – 12):** Ensuring sustainable consumption and production patterns, as a goal, aims to reduce climate change and negative environmental impacts
- **Climate action (SDG - 13):** AI can be useful in generating an alert in climatic situations. It can produce alerts and notify the relevant department about climate change, prompting action.
- **Life below water (SDG - 14):** The life below water, like animals living under the sea, under the river, plays an important role in maintaining the temperature and makes the earth suitable for all of us. AI can impact on these by recording their needs and improve the supply chain.
- **Life on Land (SDG - 15):** AI systems can be helpful for life on land as well. It can promote the use of ecosystems through platforms such as social media and search engines.
- **Peace Justice and Strong institutions (SDG - 16):** Justice and strong institutions based on the rules formed by the different groups of people. So, it is very highly impacted by AI. The tools can be trained in the laws and enforcement requirements as needed.
- **Partnerships for the goals (SDG - 17):** AI can help to provide a partnership and support the partner to achieve the goal.

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Possibilities of Artificial Intelligence technology in Health

InferVision

It is developed in China, using Artificial Intelligence in medical services. It uses deep learning and computer vision. Recently, it was used in China to fight against COVID-19. In China, it generates about 1.5 billion CT scans per year only for lung cancer. It accurately and efficiently detects cancer and helps radiologists.

DeepMind

DeepMind was created by Google and is influenced by Neuroscience. It is designed to copy the human brain's behavior. In the past, it defeated many human players in different games. Now, Google is expanding this system to the healthcare sector to reduce the time required for planning treatments and diagnosing illnesses

The possibilities of Artificial Intelligence technology by Google

Google Fishing watch

This can be used by law agencies to prevent illegal fishing. It is also a Google product that uses cloud computing, machine learning, and geospatial mapping powered by AI and Satellite Imagery to predict the location of vessels at sea. It identifies the instances of illegal fishing.

Google Brain

It was started in 2011 and used for image recognition at the initial stage. As time goes on, now used for

- Many purposes as follows:
 1. Image Enhancement
 2. Natural Language Processing
 3. YouTube video .recommendations

Possibilities of AI technology for customer experience

Walmart

It is a mall and retailer like D-mart, serving through both its offline and online stores. It is using AI technology to improve its customer experience, both in-store and online. It uses AI with IoT, Scan and Go, and pickup towers. It is also experimenting with facial recognition technology to gauge whether a customer is happy or sad while in the store.

Hello Barbie

In the market, dolls are available that can talk to humans. These toys use AI concepts like Language Processing, Machine Learning, and Advanced Analytics. These toys have a microphone that records the child's voice, selects dialogue from the server, and gives a response.

Talking with machines

It is an audio drama created by the BBC. It allows listeners to have a conversation with machines and to answer questions. By asking these questions, you add something to the story. It works with Google Echo and Google Home.

Possibilities of AI technology for vehicles

- Volvo
 - Volvo is a car company that uses AI for car servicing and the replacement of different components. It collects vehicle data, improves performance in stressful situations, and enhances the driver and passenger experience. It also used these data for research and development.
- BMW
 - 30 million raw parts handled by BMW's logistic team daily, shipped from 4500 suppliers around 31 different countries. BMW uses AI to perform some tasks and to support car manufacturing for quality control. In 2019, BMW released an open-source algorithm so that software developers can review, edit, and improve the source code for quality control. Even AI helps make decisions across car manufacturing, sales, and service. BMW has set a target that they will manufacture a self-driving car without any human intervention by 2021.
- IBM Watson
 - IBM Watson facilitates enterprise-based services, apps, and tools for developing and managing products, apps, or various corporate services. It uses many services, such as IBM Cloud, AWS, Azure, Google, or your own private cloud platform. [Click here to read more...](#)
- John Deere
 - It provides farm vehicles that use GPS and AI technology named Fram Sight for automatic plowing and sowing of the farms. This company uses AI technology in Agriculture. It checks pest presence on the crop and reports whether pesticides need to be applied.

Possibilities of Artificial Intelligence technology for Media

RADAR (Reporters and Data Robots)

It is popularly used as RADAR AI. It is a service that collects data from various sources, such as government agencies, local authorities, and public services, and publishes news. It writes around 30K stories per month and uses Natural Language Technology to generate them.

Netflix

Netflix uses AI to predict what its customers watch. This data will then be provided to content providers and content creators. Based on AI recommendations, new content will be created, helping Netflix commit to new seasons of its shows.

Cortana

Cortana is a product of Microsoft. If you have Windows 10 installed, you might have used it or seen it. It is a virtual assistant that performs various roles. For example, it can work as a chatbot on Skype and display news, weather, travel, etc.

*Thank
you*

